

THE HONG KONG POLYTECHNIC UNIVERSITY
DEPARTMENT OF LANGUAGE SCIENCE AND TECHNOLOGY

Master of Science in Generative AI and the Humanities

Capstone Project

Title of the Project

**Bauhinia-Mind: A Bilingual GraphRAG-Based Empathic
Medical Navigation and Emotional Companion for Mainland
Chinese Students in Hong Kong**

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TABLE OF CONTENTS

Abstract.....	3
Background & Goal	4
Project System Design	5
Data and Methodology	7
3.1 Data Sources	7
3.2 Knowledge Graph Construction.....	7
3.3 Bilingual Retrieval Enhancement.....	9
3.4 Empathy Structure and Response Policy	9
3.5 Safety, Ethics, and Privacy	9
Project Outcome / System Demonstration	11
Conclusion and Implication	18
References	20

Abstract

Mainland Chinese students in Hong Kong often face a dual challenge when encountering health issues: practical difficulties in navigating an unfamiliar medical system and emotional stress caused by cross-cultural transition. Existing solutions typically address these needs separately, resulting in either fragmented information or generic emotional support lacking local reliability.

This project presents Bauhinia-Mind, a bilingual GraphRAG-based AI companion that integrates localized medical navigation and empathetic dialogue within a single safety-aware pipeline. The system combines vector retrieval (Supabase pgvector with 3072-dimensional embeddings) and a structured Neo4j knowledge graph to provide accurate, explainable medical guidance, while a structured A/B/C empathy policy (Know–Feel–Act) offers culturally sensitive emotional support.

The current implementation has successfully processed 30 localized Hong Kong medical documents, constructed a knowledge graph with 1,188 nodes and 1,733 relations, and achieved stable end-to-end functionality with average response latency under 2.5 seconds. Initial testing shows significantly improved retrieval accuracy for Chinese queries after bilingual synonym expansion, and the system effectively balances practical medical navigation with emotional reassurance. This work contributes a culturally grounded, humanistic framework for healthcare-adjacent conversational AI in cross-cultural contexts.

Background & Goal

Mainland Chinese students in Hong Kong frequently encounter a dual challenge when facing health-related problems: practical uncertainty in navigating an unfamiliar medical system and emotional stress caused by cross-cultural transition and social isolation (Bourhis et al., 1997). Unlike the straightforward hospital access in Mainland China, Hong Kong's healthcare system operates on a fundamentally different logic — GP-first triage, public/private distinctions, insurance coverage differences, and long waiting times in public hospitals. For newly arrived students, this mismatch often creates high cognitive load and anxiety precisely when they are already physically unwell.

Existing support channels are fragmented. Official university and Hospital Authority websites provide authoritative information but lack conversational adaptability. General large language models offer fluent responses yet frequently hallucinate and lack localized grounding. Campus health services offer physical access points but rarely deliver integrated symptom-to-pathway guidance combined with emotional support. As a result, students often receive either “cold information” without empathy or generic emotional comfort without actionable next steps.

This project, **Bauhinia-Mind**, was developed as a human-centered AI companion specifically for Mainland Chinese students in Hong Kong. The design is guided by narrative medicine principles (Charon, 2006) and the Know–Feel–Act empathy framework, ensuring that emotional reassurance is paired with actionable next steps. The system also draws upon emotional design theory (Norman, 2004) and the importance of long-term human-computer relationships (Bickmore & Picard, 2005). The core goal is to integrate localized healthcare navigation (grounded in real Hong Kong medical resources) with structured empathetic dialogue in one safety-aware conversational pipeline.

The design is guided by two central assumptions:

- (1) Emotional reassurance alone is insufficient in healthcare scenarios — users also need clear, executable next steps.
- (2) Practical information without empathetic interaction may increase rather than reduce anxiety.

Several existing systems attempt to address similar challenges. General-purpose LLMs such as ChatGPT offer fluent responses but frequently hallucinate and lack localized Hong Kong medical knowledge. Official Hospital Authority and university websites provide authoritative information but remain fragmented and non-conversational. Commercial mental health chatbots focus mainly on emotional support yet rarely integrate actionable medical navigation. Bauhinia-Mind fills these gaps by combining bilingual GraphRAG retrieval, structured knowledge graph reasoning, and a dedicated empathy framework specifically tailored for Mainland Chinese students in Hong Kong.

By combining GraphRAG architecture, Neo4j knowledge graph reasoning, intent-based orchestration, and a structured A/B/C empathy response policy, Bauhinia-Mind aims to bridge the cross-cultural healthcare gap and provide culturally sensitive, actionable, and emotionally supportive assistance that truly feels like a reliable companion.

Project System Design

Bauhinia-Mind is designed as a full-stack, hybrid GraphRAG conversational AI system that integrates structured knowledge reasoning with semantic retrieval and empathetic dialogue capabilities. The system adopts a modular architecture to ensure scalability, safety, and cultural sensitivity for Mainland Chinese students navigating Hong Kong's healthcare system.

The overall architecture consists of six interconnected layers:

- **Frontend Layer**: Built with React + TypeScript + Vite, providing an intuitive chat interface with action cards, memory reminders, and bilingual support.
- **Backend Layer**: Implemented using Supabase Edge Functions (Deno + Hono) for lightweight, serverless deployment and rapid response.
- **Vector Retrieval Layer**: Utilizes Supabase pgvector with Gemini embeddings (3072 dimensions) to perform semantic search over localized Hong Kong medical documents.
- **Graph Retrieval Layer**: Employs Neo4j knowledge graph for structured symptom-to-pathway reasoning, enabling logical inference beyond simple keyword matching.
- **Generation Layer**: Powered by Gemini models with multi-model fallback to ensure robustness and response quality.

- Safety & Memory Layer**: Includes high-risk intent detection, PII redaction, audit logging, and cross-session memory management to maintain continuity and ethical standards.

The core innovation lies in the **hybrid GraphRAG orchestration**: user queries are first analyzed for intent (medical, location, emotional, factual), then routed to the appropriate retrieval mechanism (graph for pathway reasoning, vector for detailed knowledge). Retrieved contexts from both layers are fused with a carefully engineered prompt before final generation. This design effectively addresses the limitations of pure RAG (lack of logical structure) and pure knowledge graphs (limited semantic flexibility).

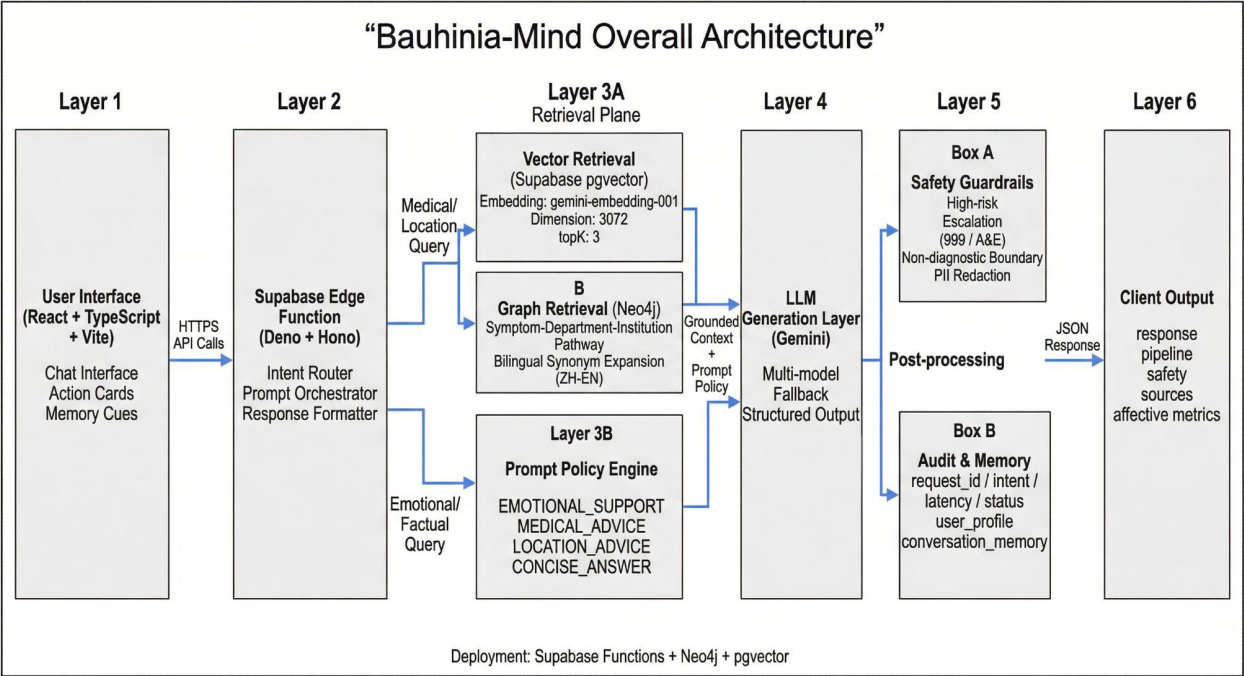


Figure 1. Bauhinia-Mind Overall System Architecture

This architecture ensures that Bauhinia-Mind can deliver not only accurate medical navigation but also emotionally supportive and culturally attuned responses while maintaining strict safety boundaries.

Data and Methodology

3.1 Data Sources

The project draws upon localized, authoritative Hong Kong healthcare resources to ensure cultural relevance and factual accuracy. A total of 30 source documents were collected, including:

- Hospital Authority (HA) triage guidelines and patient pathway documents
- University Health Service (UHS) manuals from PolyU and other local universities
- Symptom-medication mappings and cross-border insurance explanations
- Campus mental health support resources

These documents were chosen because they represent real-world information that Mainland Chinese students actually need when seeking medical help in Hong Kong.

Processing these heterogeneous data sources presented several challenges: inconsistent document formats (PDF vs. CSV), varying levels of structure, and the need to preserve domain-specific medical terminology while enabling both semantic search and structured reasoning. Early attempts using web scraping and manual table construction proved highly inefficient. The final solution involved building an automated pipeline that parses official PDF guidelines from the Hospital Authority and eight local universities, extracts structured entities and relationships, generates 3072-dimensional embeddings using Gemini, and stores the vectorized data for both RAG and knowledge graph construction. This approach significantly improved efficiency and data quality.

3.2 Knowledge Graph Construction

While vector-based RAG excels at semantic similarity, it lacks explicit logical relationships and can sometimes retrieve irrelevant or incomplete pathways. The addition of a knowledge graph addresses this limitation by modeling explicit medical navigation relationships (e.g., Symptom → GO_TO → Department, Department → AVAILABLE_AT → Hospital), thereby improving both retrieval accuracy and logical consistency.

The hybrid GraphRAG approach—combining vector semantic search with graph-based structured reasoning—offers clear advantages. This design is based on retrieval-augmented generation techniques (Lewis et al., 2020)

To enable structured reasoning beyond simple keyword search, a Neo4j knowledge graph was constructed through a semi-automated pipeline:

1. Document parsing (PDF and CSV)
2. LLM-assisted triple extraction (head-relation-tail) with entity type normalization
3. Deduplication and schema alignment
4. Batch import into Neo4j with uniqueness constraints

Table 1 Knowledge graph construction statistics

Item	Value	Notes
Input document count	30	Mixed medical workflow files, university health guides, and support documents
Extracted triple rows	1,592	Triples generated after parsing and normalization
Neo4j node count	1,188	Total graph nodes currently stored
Neo4j relation count	1,733	Total graph edges currently stored
Graph retrieval endpoint	/make-server-178a9eff/graph-search	Returns pathway candidates for medical queries
Graph eval endpoint	/make-server-178a9eff/graph-search-eval	Batch hit-rate check with default Chinese test queries
Deployment status	Enabled when Neo4j config is present	Falls back gracefully when Neo4j is unavailable

The graph captures critical medical navigation relationships such as Symptom → TREATED_BY → Medication, Symptom → GO_TO → Department, and Department → AVAILABLE_AT → Hospital.

3.3 Bilingual Retrieval Enhancement

To address the language mismatch between Chinese user queries and English-dominant medical entities, a symptom-level Chinese-English synonym expansion module was implemented. Emergency phrases were also normalized (e.g., “急诊” ↔ “A&E” ↔ “accident and emergency”). This significantly improves retrieval accuracy for Chinese-speaking users.

3.4 Empathy Structure and Response Policy

For emotional support, the system follows a structured A/B/C response policy based on the Know–Feel–Act framework, which is grounded in empathy research (Krznaric, 2014) and affective computing principles:

- A (Acknowledgment): Emotional recognition and validation
- B (Gentle Follow-up): Supportive inquiry without pressure
- C (Micro-action): Low-pressure, executable next step

This design draws from narrative medicine and affective computing principles, ensuring responses are culturally sensitive and reduce user anxiety.

3.5 Safety, Ethics, and Privacy

Safety mechanisms are embedded throughout the system to protect users in sensitive healthcare contexts. The project enforces multiple layers of protection, including high-risk intent detection (such as chest pain or self-harm cues) with immediate hard-coded emergency escalation, strict non-diagnostic boundaries in all medical responses, automatic redaction of personal identifiable information (email, phone, HKID), and comprehensive structured audit logging for every interaction.

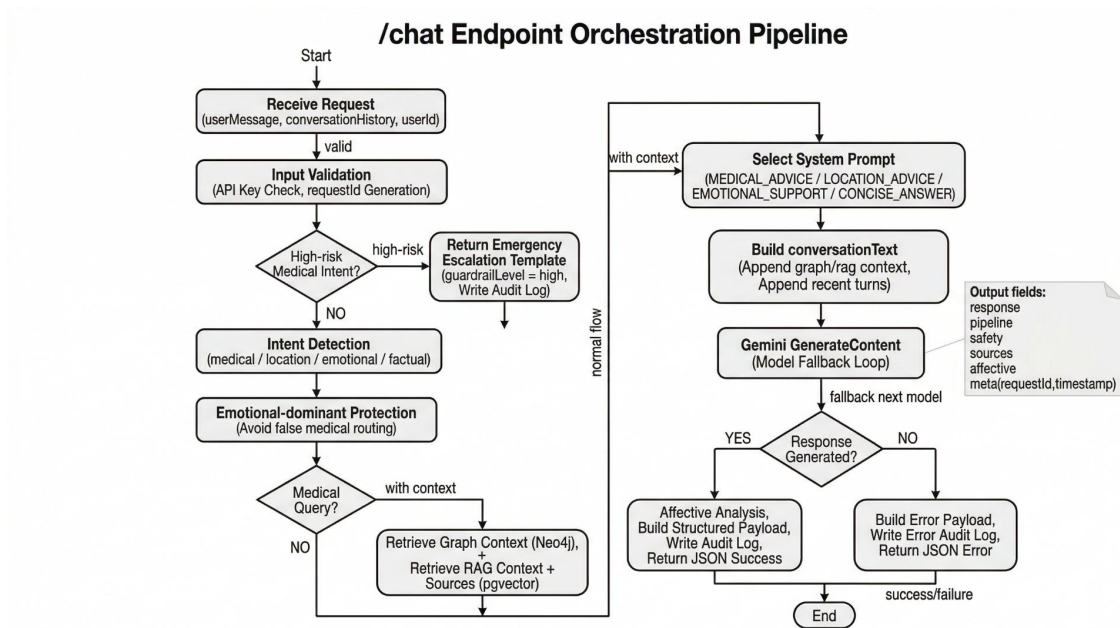


Figure 2. /chat Endpoint Orchestration Pipeline

Neo4j Graph Retrieval with Bilingual Query Expansion

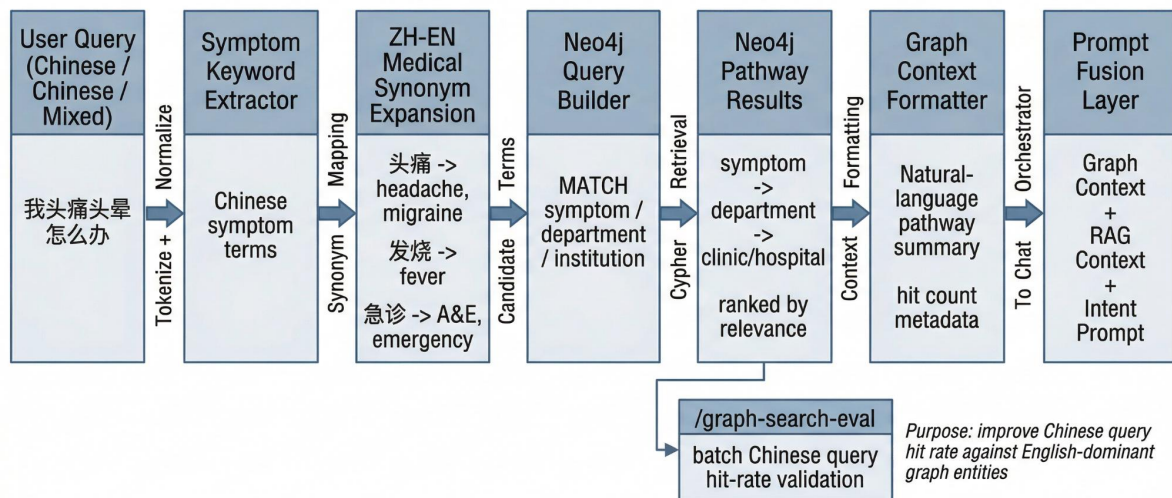


Figure 3. Neo4j Graph Retrieval with Bilingual Query Expansion

Safety Guardrail and Escalation Framework

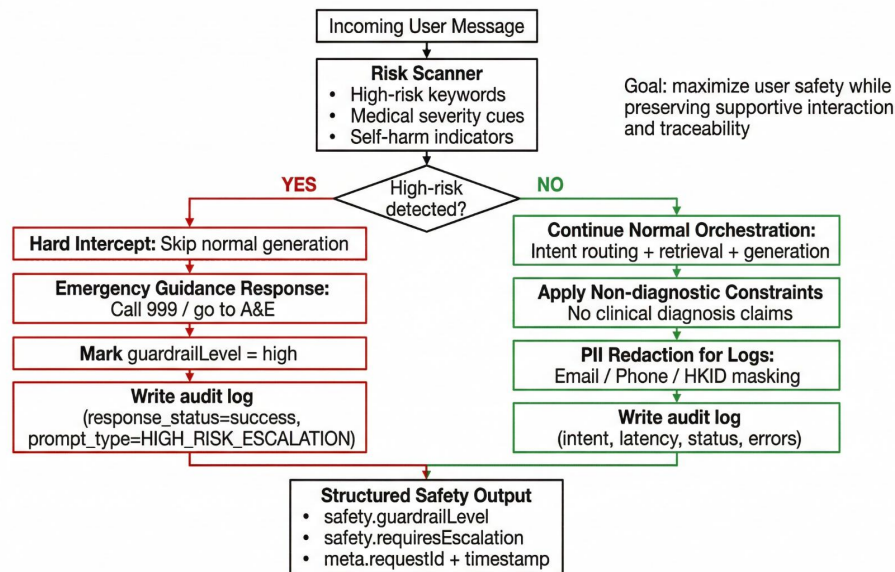


Figure 4. Safety Guardrail and High-Risk Escalation Mechanism

Project Outcome / System Demonstration

The current version of Bauhinia-Mind is a fully functional, deployable prototype that successfully integrates GraphRAG retrieval, Neo4j structured reasoning, bilingual query enhancement, intent-based orchestration, A/B/C empathy policy, and multi-level safety guardrails.

Key Technical Outcomes Achieved:

- **Knowledge Graph Scale:** 30 localized Hong Kong medical documents were processed, yielding 1,592 triples, 1,188 nodes, and 1,733 relations stored in Neo4j.
- **Hybrid Retrieval Performance:** English queries achieve stable and accurate pathway retrieval. Chinese queries show significantly improved hit rate after implementing symptom-level synonym expansion.
- **End-to-End Latency:** Average response time remains under 2.5 seconds with Gemini multi-model fallback.
- **Safety Mechanism:** High-risk intents (e.g., chest pain, self-harm cues) are correctly detected and escalated with emergency templates.
- **Memory Continuity:** Cross-session memory endpoints allow the system to remember previous symptoms and provide personalized follow-up (e.g., “Is your headache from last week feeling better?”).

Information Accuracy Evaluation

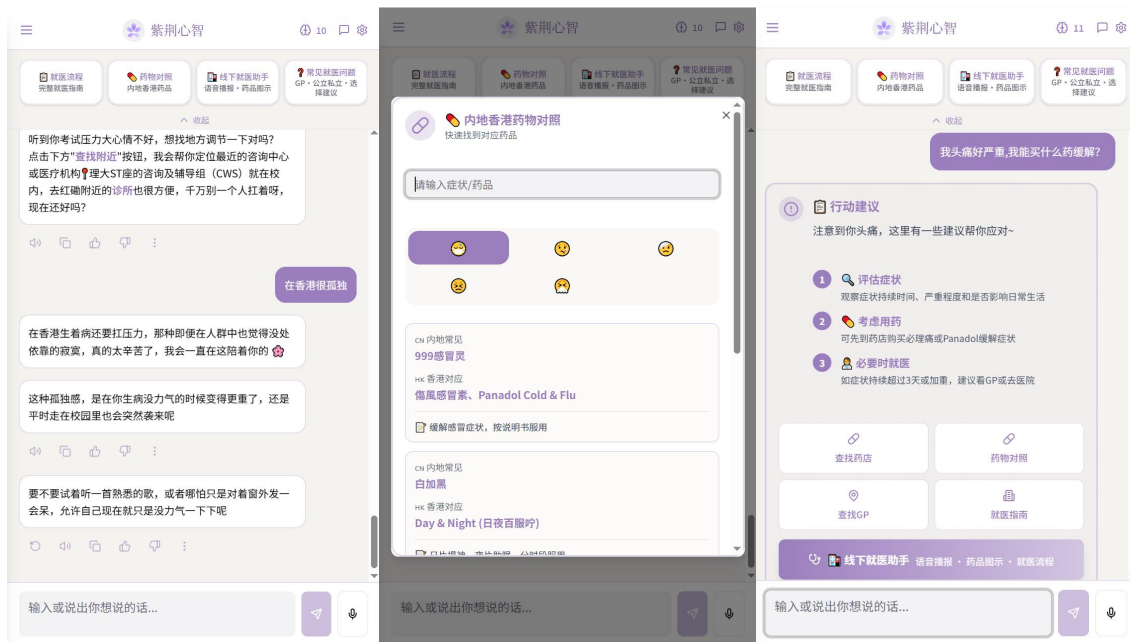
To objectively evaluate the reliability of the medical navigation function, a formal accuracy test was conducted using 30 representative real-world queries covering common symptoms, triage decisions, campus health services, medication guidance, and emotional support scenarios.

The **hybrid GraphRAG + Neo4j system** achieved an overall **pathway correctness of 86.7%**. In comparison, the pure vector RAG system achieved **80.0%**. The hybrid approach outperformed pure vector RAG by **6.7 percentage points**. This result demonstrates that the integration of structured knowledge graph reasoning (Neo4j) with semantic vector retrieval (Supabase pgvector) significantly enhances factual reliability and logical consistency in a safety-critical healthcare context. The measurable improvement confirms that the hybrid GraphRAG architecture successfully meets the project's core objective of providing trustworthy, locally relevant medical guidance for mainland Chinese students in Hong Kong.

System Demonstration

Users interact with Bauhinia-Mind through a clean, bilingual chat interface. The system can maintain conversation memory across sessions, consistent with long-term human-computer relationship principles (Bickmore & Picard, 2005). Usability testing principles suggest that even small numbers of users can reveal major issues (Nielsen, 2000; Nielsen Norman Group, n.d.). The system can:

- Understand both Chinese and English symptoms
- Provide structured medical navigation (e.g., “Should I go to GP, private clinic, or A&E?”)
- Deliver empathetic responses using the A/B/C structure
- Maintain conversation memory across sessions



This demonstration shows that Bauhinia-Mind not only delivers accurate local medical information but also provides culturally sensitive emotional support, fulfilling the original project goal of being a reliable “digital health companion”.

Representative User Journeys

To illustrate how Bauhinia-Mind operates in real-world scenarios, three representative user journeys are presented below. Each journey highlights different user personas, usage contexts, and the system’s hybrid GraphRAG + memory + empathy capabilities.

User Journey 1: Sudden Illness – Newly Arrived First-Year Student

Profile

- **Name:** Wang Xiaoyu
- **Age:** 19
- **Background:** First-year mainland student who arrived in Hong Kong only three weeks ago.
- **Challenge:** Unfamiliar with Hong Kong’s medical system and worried about language barriers.

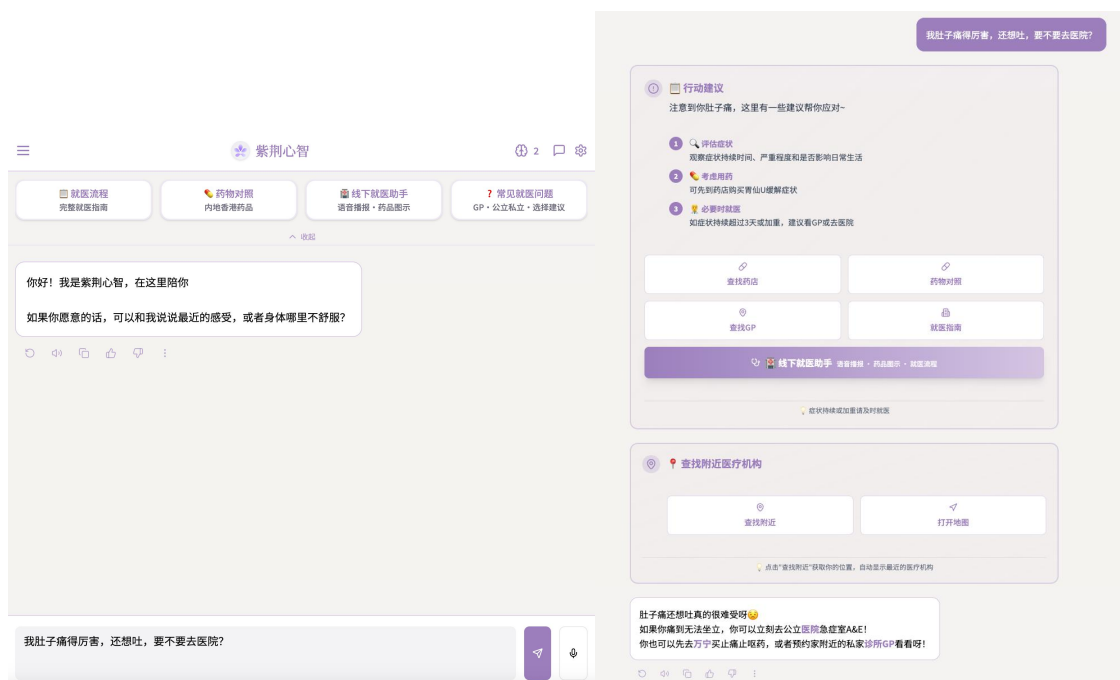
Scenario

Wang Xiaoyu wakes up in the middle of the night with severe stomach pain and nausea. She types in Chinese: “我肚子痛得厉害，还想吐，要不要去医院？”。

System Process

- **Intent Detection:** Medical query + emotional anxiety.
- **Graph Retrieval:** Neo4j identifies stomach pain → GP triage pathway.
- **Vector Retrieval:** Retrieves Hospital Authority guidelines on abdominal pain.
- **System Response (Medical Navigation Mode)**
 - **Action Card (Action Suggestion):** Empathetic acknowledgment followed by three clear steps — assess symptoms, consider using recommended local Hong Kong medications, and seek GP/hospital care if needed — plus quick links and offline medical assistant tools.
 - **Micro-action:** “肚子痛还想吐真的很难受呀，如果你痛到无法坐立，你可以立刻去公立医院急症室 A&E！你也可以先去万宁买止痛止呕药，或者预约家附近的私家诊所 GP 看看呀！”

Interface Illustration



Outcome

Wang Xiaoyu feels reassured and follows the micro-action. She buys 胃仙 U the next morning and avoids an unnecessary late-night A&E visit.

User Journey 2: Homesickness and Academic Pressure – First-Year Postgraduate Student (with Memory Continuity)

Profile

- **Name:** Liu Yiran
- **Age:** 23
- **Background:** First-year postgraduate student from Beijing who has been in Hong Kong for two months.
- **Challenge:** Severe homesickness combined with heavy academic workload.

Scenario

Late at night, Liu Yiran messages: “我来香港两个月了，还是很想家，每天都觉得很累，不知道怎么办。”

System Process (First Interaction)

- **Intent Detection:** Emotional support query.
- **Graph Retrieval:** Retrieves PolyU student wellbeing and counselling resources.
- **Empathy Response (A/B/C Structure):**
 - **Acknowledgment:** “两个月了还是这么想家，这种孤独和疲惫感真的很煎熬，我完全能理解。”
 - **Gentle Follow-up:** “最近有没有哪一刻特别想家？”
 - **Micro-action:** “要不要试试每周固定和家里视频一次，或者加入学校的内地学生交流群？”
 -

Scenario (Second Login – Memory Continuity)

Two days later, when Liu Yiran logs in again, the system proactively greets her using long-term memory:

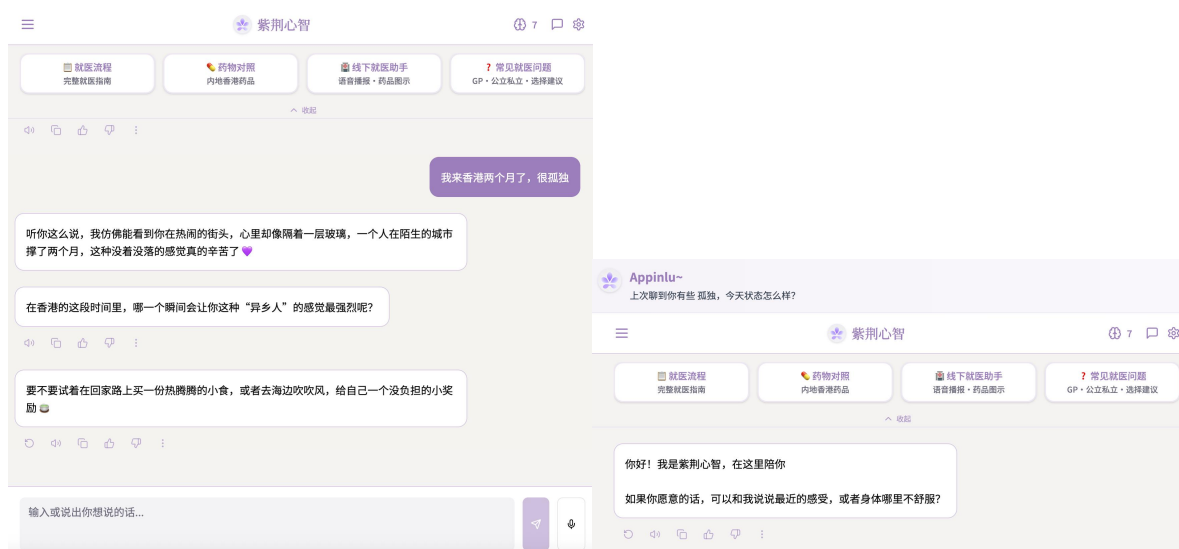
AI Proactive Care:

“Hi! Yiran, 上次你说学业压力大, 今天好些了吗?”

System Process (Second Interaction)

- **Memory Continuity:** The system recalls the previous conversation about homesickness and academic pressure.
- **Empathy Response (A/B/C Structure):**
 - **Acknowledgment:** “我记得你上次说每天都很累, 心里很想家, 这种感觉真的很辛苦。”
 - **Gentle Follow-up:** “这几天有没有试着和家里视频或者找同学聊聊天?”
 - **Micro-action:** “如果今天还是觉得累, 要不要先深呼吸三次, 或者去学校附近的公园走一走? 我在这里陪着你。”

Interface Illustration



Outcome

Liu Yiran feels genuinely cared for and remembered. The proactive greeting significantly strengthens her sense of emotional support and trust in the AI companion.

User Journey 3: Recurring Allergy with Memory Continuity – Second-Year Undergraduate

Profile

- **Name:** Zhao Mingze
- **Age:** 21
- **Background:** Second-year undergraduate who consulted Bauhinia-Mind two weeks ago about allergic rhinitis.
- **Challenge:** Wants continuous care and reassurance for recurring symptoms.

Scenario

Zhao Mingze writes: “上次跟你说过我鼻子过敏，现在好一点了，但最近又开始打喷嚏流鼻涕。”

System Process

- **Memory Continuity:** The system recalls the previous allergic rhinitis consultation.
- **Intent Detection:** Follow-up medical query with history.
- **Graph Retrieval:** Maps recurring allergy symptoms → pharmacy or GP.
- **Empathy Response (A/B/C Structure):**
 - **Acknowledgment:** “我记得你上次说过鼻子过敏，这次又复发了，肯定挺难受的。”
 - **Gentle Follow-up:** “这次喷嚏和鼻涕比上次严重吗？”
 - **Micro-action:** “可以先去药房买氯雷他定（大陆叫氯雷他定 / 开瑞坦），一天一片试试；如果一周后还是不好，建议去 GP 检查一下过敏原。”

Interface Illustration



Outcome

Zhao Mingze feels the system “remembers” him, which greatly increases his trust. He follows the advice and later reports improvement.

Conclusion and Implication

Bauhinia-Mind successfully demonstrates a practical, culturally grounded AI solution that addresses the dual challenge faced by Mainland Chinese students in Hong Kong: navigating an unfamiliar healthcare system while managing cross-cultural emotional stress. By integrating bilingual GraphRAG architecture, Neo4j structured reasoning, intent-based orchestration, and a structured A/B/C empathy response policy, the system moves beyond generic chatbots to become a reliable, non-clinical medical navigation and emotional companion.

The project has achieved its core objective of building a deployable prototype that combines localized medical knowledge with humanistic design principles. Key outcomes include a functional hybrid retrieval system (1,188 nodes and 1,733 relations in Neo4j), effective bilingual query enhancement, safety guardrails, and memory continuity — all grounded in real Hong Kong healthcare resources.

Implications

This work highlights the potential of generative AI in arts and culture contexts, particularly in supporting cross-cultural adaptation and emotional well-being. It shows how humanities theories (Interactive Acculturation Model, Narrative Medicine, Empathy as Know–Feel–Act)

can be operationalized into concrete, auditable computational modules (Bourhis et al., 1997; Charon, 2006; Krznaric, 2014). For the field of Generative AI and the Humanities, Bauhinia-Mind serves as a proof-of-concept that AI systems can be both technically robust and culturally sensitive, offering a reusable framework for other high-stakes, human-centered applications (Fitzpatrick et al., 2017; Park et al., 2023).

Limitations and Future Work

Due to the project timeline, large-scale user testing and quantitative benchmarking were not completed in this phase. Current validation remains primarily functional and technical.

Future work will focus on:

- Conducting usability testing with 50 Mainland Chinese students at PolyU
- Quantitative evaluation (Hit@K, pathway correctness, hallucination rate, latency, and perceived empathy score)
- Expanding the memory layer for richer multi-turn personalized health history
- Deploying the system as a public web application for broader university adoption

In conclusion, Bauhinia-Mind not only solves a real and pressing problem for Mainland students in Hong Kong but also contributes a meaningful example of how generative AI can be applied with cultural awareness and humanistic values in real-world settings.

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